1. Basic Search

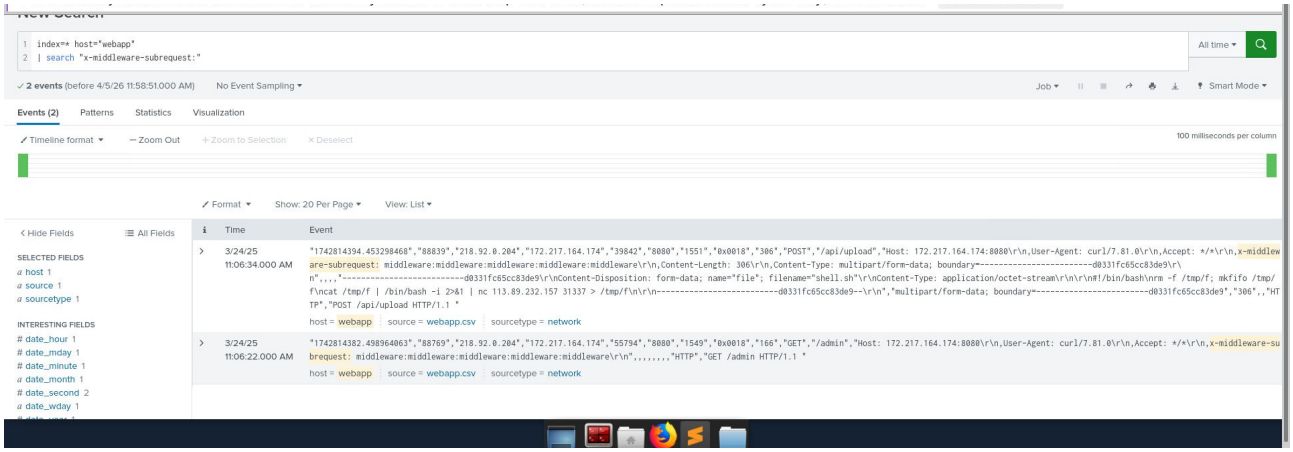
index=* "x-middleware-subrequest:"

Finds any occurrence of the header across logs

2. Search Web Traffic

index=* host="webapp"

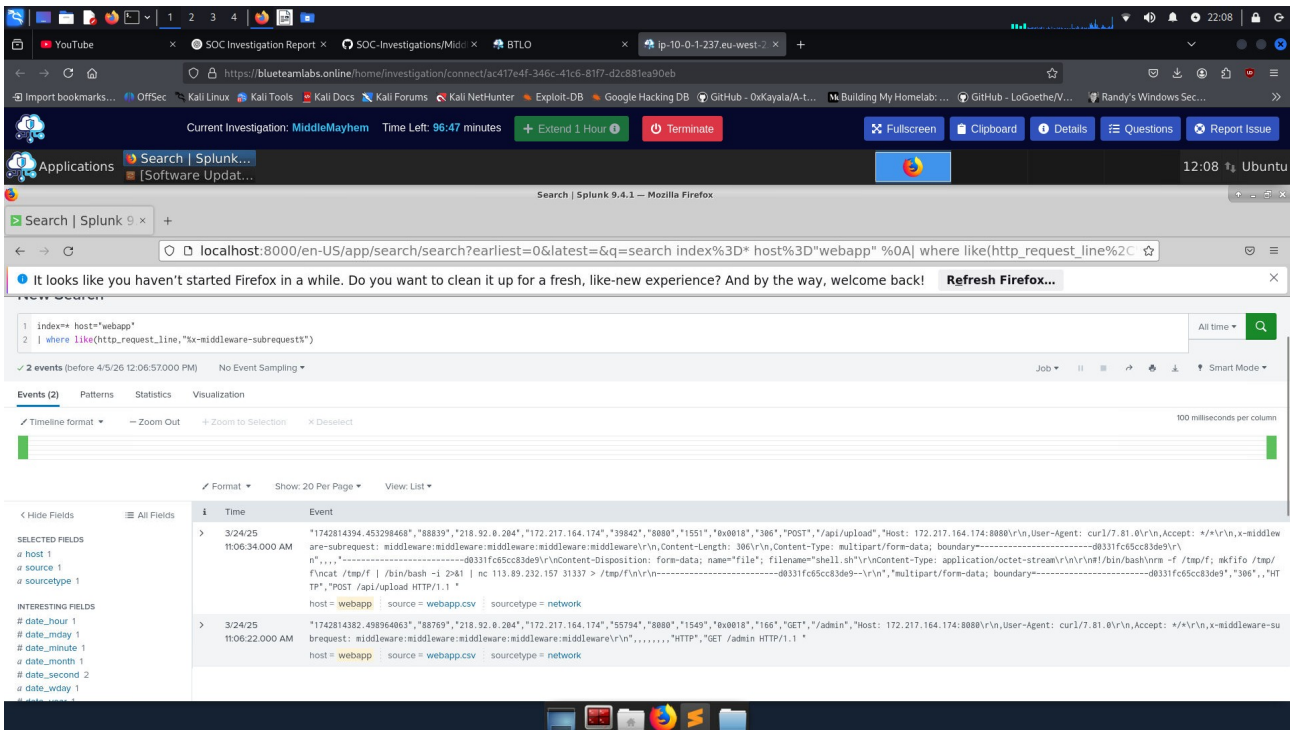
| search "x-middleware-subrequest:"



Filters to web-related logs only

3. Structured Field Search

index=* sourcetype=web
| where like(http_header, "%x-middleware-subrequest%")



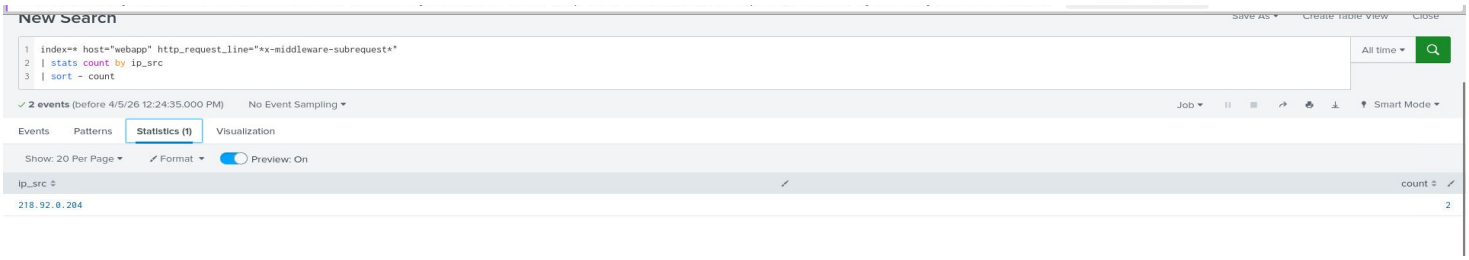
4. External Attacker Detection

index=* sourcetype=web http_request_line="*x-middleware-subrequest*"
| where NOT cidrmatch("10.0.0.0/8", ip_src)
  AND NOT cidrmatch("172.16.0.0/12", ip_src)
  AND NOT cidrmatch("192.168.0.0/16", ip_src)

Searches for x-middleware-subrequest only from external IP addresses. Detects external IPs abusing internal headers.

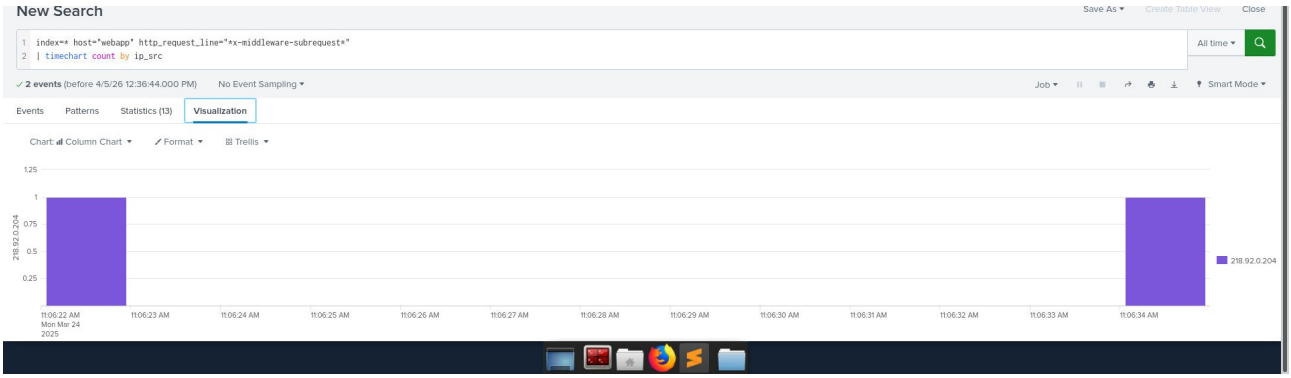
5. Find Top Attackers

index=* sourcetype=web http_header="*x-middleware-subrequest*"
| stats count by ip_src
| sort - count



6. Timeline of Exploitation

index=* sourcetype=web http_header="*x-middleware-subrequest*"
| timechart count by ip_src



Shows the times of attack or usage of internal headers.

7. Correlate with Sensitive Endpoints

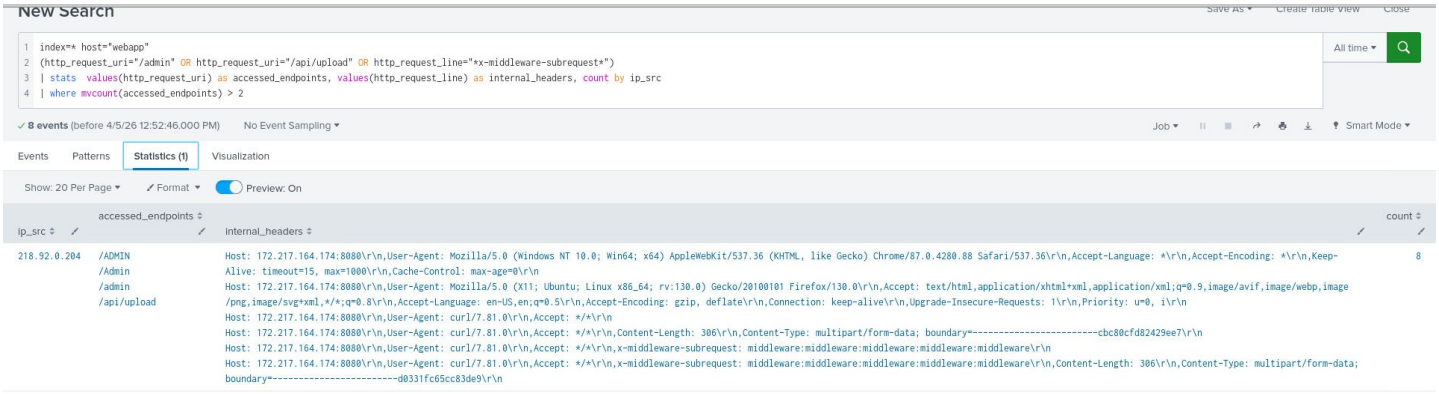
index=* host="webapp" http_request_line="*x-middleware-subrequest*"
| search http_request_uri="/admin" OR http_request_uri="/api/upload"

| stats count by ip_src,http_request_uri

Provides detail of Header abuse with specific endpoints usage.

8. Full Attack Chain Detection

index=* host="webapp"
(http_request_uri="/admin" OR http_request_uri="/api/upload" OR http_request_line="*x-middleware-subrequest*")
| stats values(http_request_uri) as accessed_endpoints, values(http_request_line) as internal_headers, count by ip_src
| where mvcount(accessed_endpoints) > 2



This SPL provides an end-to-end flow of attack starting from Recon → exploit → post-exploitation behavior